EV Technology Course — Module 1: EV Vehicle Architecture and System Overview

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An EV is eight subsystems working as one: energy storage, propulsion, charging, power distribution, thermal management, control, chassis and body, and low-voltage electronics. This module maps them, traces power and signals through a car, a scooter and a bus, and explains why purpose-built "skateboard" platforms are replacing EVs converted from petrol cars.

1. Learning objectives

- Name the eight subsystems of an EV and what each one does.
- Trace the high-voltage power path and the control signals from charging inlet to wheel.
- Compare EV architectures for two-wheelers, cars and buses.
- Explain FWD, RWD and AWD layouts, and converted vs dedicated platforms.
- Calculate motor speed, wheel torque, tractive force and gradeability.

2. Fundamentals

A petrol car is organised around the engine: everything else (gearbox, exhaust, fuel tank, radiator) serves it. An EV is organised around the battery and the flow of electrical energy. Three ideas hold the whole architecture together:

1. **Two voltage worlds.** A high-voltage (HV) system, typically 300–800 V DC in cars, moves power. A low-voltage (LV) system, 12 V (sometimes 48 V), runs the electronics, lights and safety systems.
2. **Power converters everywhere.** The battery stores DC, the motor needs variable AC, the grid gives AC, and the electronics need 12 V. Each conversion is done by a dedicated power-electronic box.
3. **Software coordinates everything.** The vehicle control unit (VCU), battery management system (BMS) and motor controller (MCU) agree every few milliseconds how much power can flow, in which direction.

3. The eight subsystems

Subsystem	Main parts	Job	Covered in
Energy storage	Battery pack, BMS	Store and release energy safely	Battery course
Propulsion (e-drive)	Inverter, motor, reduction gear, differential	Turn DC into wheel torque; recover energy on braking	Modules 2–4
Charging	Inlet, onboard charger (OBC), charge controller	Take energy from the grid	Battery course Module 9; Module 6
Power distribution	HV junction box, contactors, fuses, HV harness, DC-DC converter	Route and protect HV; feed the 12 V system	Module 6
Thermal management	Pumps, valves, chiller, heat pump, radiator, PTC heater	Keep battery, e-drive and cabin at the right temperature	Module 7
Control	VCU, MCU, BMS, brake/ESC controller, gateway	Decide and coordinate torque, limits and modes	Modules 5, 10
Chassis and body	Skateboard frame, crash structures, suspension, brakes, steering	Carry loads, protect the battery and occupants	Modules 8–9
LV electronics	12 V battery, ECUs, lights, infotainment, ADAS	Everything that is not traction	Modules 6, 10

Architectures by vehicle type

Vehicle	Typical HV level	Battery	Motor and drive	Notes
E-scooter / e-motorcycle	48–80 V (below or near 60 V DC)	2–5 kWh	BLDC hub motor or frame-mounted PMSM with belt/chain	Often no separate VCU; controller does most functions

Vehicle	Typical HV level	Battery	Motor and drive	Notes
E-rickshaw / e-auto (3W)	48–60 V	5–10 kWh	Motor + fixed gear + differential on rear axle	Lead-acid still common in low-end e-rickshaws; LFP in L5 autos
Passenger car	300–800 V	30–100+ kWh	One or two e-axles	Liquid-cooled battery and e-drive
City bus	600–750 V	250–400 kWh	Central motor or e-axle / hub motors	Depot charging; long duty cycles

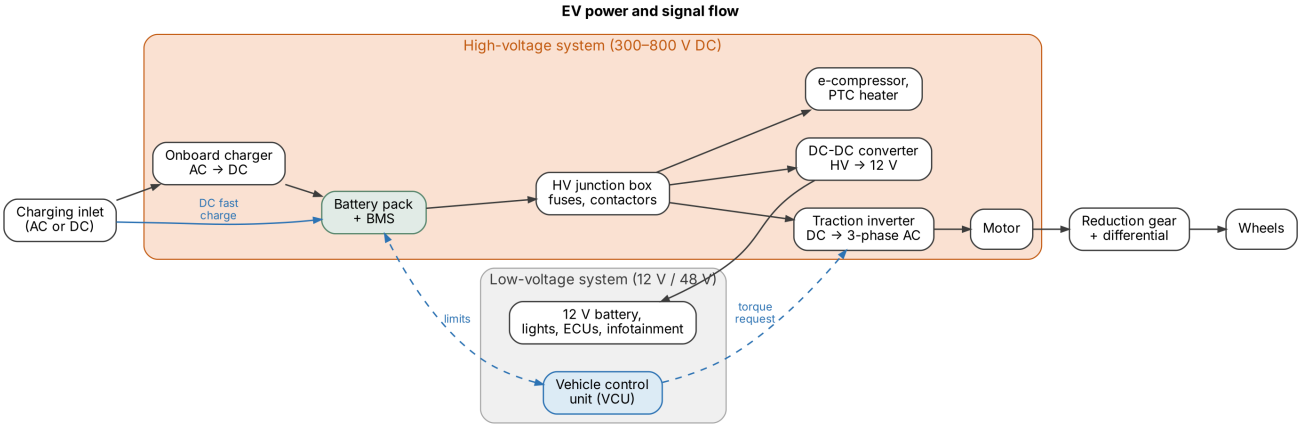
Drive layouts

- **FWD (front motor):** common when converting a petrol front-wheel-drive car (e.g. Tata Nexon EV).
- **RWD (rear motor):** better traction under acceleration, tighter turning circle; common on dedicated platforms (e.g. Mahindra BE 6, Tesla Model 3 RWD).
- **AWD (dual motor):** one e-axle per axle; enables torque vectoring and redundancy.
- **Tri- or quad-motor:** high-performance models; one motor per rear wheel (or per wheel) allows precise torque vectoring.

Converted vs dedicated platforms

	Converted (ICE-derived)	Dedicated ("skateboard")
Battery space	Squeezed into tunnel, floor and boot	Flat pack between the axles
Interior	Same as petrol car	Flat floor, longer wheelbase, more cabin space
Cost and time	Lower investment, faster launch	Higher upfront cost, better long-term economics
Examples	Tata Nexon EV, Tiago EV, MG ZS EV	Mahindra INGLO (BE 6, XEV 9e), Tata acti.ev (Punch EV, Curvv EV), Hyundai E-GMP

4. Diagram: power and signal flow



Orange is the high-voltage system, grey the 12 V system. Solid arrows carry power; dashed blue arrows carry control signals.

5. Step-by-step: from "start" to driving, braking and charging

1. **Wake-up:** pressing start powers the 12 V network; the VCU, BMS and MCU boot.
2. **Safety checks:** the BMS checks cell voltages, temperatures, insulation resistance and the HV interlock loop.
3. **HV on:** the BMS closes the negative contactor, pre-charges the inverter's DC-link capacitor, then closes the positive contactor (battery course Module 6). The dashboard shows "Ready".
4. **Drive:** the accelerator pedal sends a torque request to the VCU. The VCU checks BMS power limits, traction and drive mode, then sends a torque command to the MCU, which drives the inverter.
5. **Brake:** lifting off or pressing the brake triggers regenerative braking; the motor becomes a generator and the inverter sends DC back to the battery. Friction brakes add whatever regen cannot supply.
6. **Park and charge:** contactors open when parked. Plugging in wakes the charge controller, which talks to the charger and the BMS; AC goes through the OBC, DC goes straight to the pack.
7. **Always on:** the DC-DC converter keeps the 12 V battery charged whenever HV is live.

6. Engineering equations

Vehicle speed from motor speed:

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v = \frac{n_{motor}}{i_g} \times \frac{2\pi r_w}{60}
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Wheel torque and tractive force:

$$T_{wheel} = T_{motor} \times i_g \times \eta_g \qquad F_t = \frac{T_{wheel}}{r_w}$$

Force needed (acceleration, rolling, aero, grade):

$$F = m\,a + m\,g\,C_{rr}\cos\theta + \tfrac{1}{2}\rho\,C_d\,A\,v^2 + m\,g\sin\theta$$

Power:

$$P = F \times v \qquad P = T \times \omega = T \times \frac{2\pi n}{60}$$

v in m/s, n in rpm, i_g = gear ratio, r_w = wheel rolling radius (m), η_g = gear efficiency, T in N·m, F in N, m in kg, θ = road angle (grade % = 100 × tan θ), P in W.

7. Worked examples

Example 1: motor speed at 120 km/h. r_w = 0.33 m, i_g = 9.

- v = 33.33 m/s → wheel speed = 33.33 ÷ (2π × 0.33) × 60 = **965 rpm** → motor = 965 × 9 = **8,681 rpm**.

Example 2: launch force. Motor 250 N·m, i_g = 9, η_g = 0.95, r_w = 0.33 m, mass 1,700 kg.

- Wheel torque = 250 × 9 × 0.95 = **2,138 N·m** → F_t = 2,138 ÷ 0.33 = **6,477 N** → a ≈ 6,477 ÷ 1,700 = **3.8 m/s²** before losses and traction limits (about 0.39 g).

Example 3: gradeability. Force to hold 1,700 kg on a 20% grade (θ = 11.3°): 1,700 × 9.81 × sin 11.3° = **3,271 N**, well within the 6,477 N available. This is why EVs pull strongly up hills from standstill.

8. Real-world EV examples

- **Tata Nexon EV:** converted platform; front motor, FWD; battery under floor and tunnel.
- **Mahindra BE 6 / XEV 9e:** dedicated INGLO platform; rear motor (RWD) or dual motor (AWD).
- **Tesla Model 3/Y:** dedicated platform; rear motor or dual-motor AWD; highly integrated electronics.
- **Ather 450:** frame-mounted motor with belt drive; many low-cost scooters use hub motors.
- **Electric buses (Tata, Olectra, PMI):** large packs on roof and/or floor; motor driving a rear axle.

9. Manufacturing applications

- An EV deletes engine, gearbox, fuel and exhaust lines but adds HV battery installation ("marriage"), HV harness routing, coolant filling of extra loops, insulation and functional tests.
- Battery marriage is a heavy, precise operation: 300–600 kg lifted from below and torqued to the body, often with automated nut-runners.
- Converted platforms can share the body shop and paint shop with the petrol model; dedicated platforms often need new body tooling but simplify final assembly.
- Costing view: the battery and e-drive dominate the BOM (Module 13); platform choice decides how much carry-over you can reuse.

10. Common mistakes and misconceptions

- **"EVs have no gearbox."** They have a single-speed reduction gear and a differential.
- **"The battery drives the motor directly."** The inverter sits in between and controls everything.
- **"12 V systems are obsolete in EVs."** Every EV still needs LV for ECUs, lights and safety systems.
- **"AWD is only for performance."** It also adds redundancy and stability, and can improve efficiency when one motor disengages.

11. Practical case study: converted vs dedicated platform

Problem. An Indian OEM must launch an electric compact SUV in 18 months and a long-range model in 4 years.

Analysis. Converting the petrol SUV is fast and uses existing tooling, but the battery must fit around the tunnel and fuel-tank area, limiting capacity to about 40–45 kWh and compromising the floor. A dedicated platform takes longer but allows 60–80 kWh, a flat floor and better crash protection.

Decision. Convert for the first launch; build a dedicated skateboard for the second generation, designed to share battery modules and e-drive across several bodies.

Lesson. This is the path Tata (Nexon EV → acti.ev) and Mahindra (eXUV400 → INGLO) actually followed.

12. Revision notes

- Eight subsystems: storage, propulsion, charging, distribution, thermal, control, chassis/body, LV electronics.
- HV moves power; LV runs electronics; converters link them.
- VCU arbitrates torque; BMS sets limits; MCU executes.
- $v = (n \div i) \times 2\pi r \div 60$; $T_{wheel} = T_{motor} \times i \times \eta$.

13. Glossary

Term	Meaning
VCU	Vehicle control unit; top-level controller
MCU	Motor control unit (inside the inverter)
OBC	Onboard charger
PDU / junction box	HV power distribution unit
Skateboard	Flat chassis with battery between the axles
e-axle	Motor, inverter and gearbox combined on an axle
Gradeability	Steepest slope a vehicle can climb or start on

14. Knowledge check

Conceptual

1. Why does an EV need both a high-voltage and a low-voltage system?
2. What does the VCU do that the BMS and MCU cannot do alone?
3. Why do many dedicated EV platforms use rear-wheel drive?
4. What are the main compromises of converting a petrol platform to electric?
5. Why does an EV still need a differential?

Numerical

1. Motor maximum speed 16,000 rpm, ratio 9.5, $r_w = 0.32$ m. What is the top speed?
2. A scooter motor gives 25 N·m through an 8:1 reduction ($\eta = 0.92$) to a 0.25 m wheel. Force at the wheel, and acceleration for 150 kg (bike + rider)?
3. Power needed to climb a 10% grade at 40 km/h with 1,800 kg (ignore aero and rolling)?

Application

1. You are costing an ICE-to-EV conversion. List parts deleted and parts added.
2. Recommend a drive layout for an Indian electric taxi and justify it.

15. Answer key

Conceptual

1. HV moves tens of kilowatts efficiently; LV runs standard, safe, low-power electronics and keeps safety systems alive even when HV is off.
2. It arbitrates between driver request, BMS limits, traction/stability control, drive modes and thermal limits, and coordinates contactors and charging.
3. Under acceleration weight shifts rearward, improving traction; no driveshaft packaging problem; better steering lock.
4. Smaller, oddly shaped battery; raised floor; compromised crash paths; cooling and wiring routed around ICE-era structures.
5. Inner and outer wheels travel different distances in a turn; the differential lets them turn at different speeds.

Numerical

1. Wheel rpm = 16,000 ÷ 9.5 = 1,684 → v = 1,684 × 2π × 0.32 ÷ 60 = 56.4 m/s = **203 km/h** (other limits usually cap it lower).
2. F = 25 × 8 × 0.92 ÷ 0.25 = **736 N** → a = 736 ÷ 150 = **4.9 m/s²** before losses.
3. F = 1,800 × 9.81 × sin(5.71°) = 1,757 N; P = 1,757 × 11.11 = **19.5 kW**.

Application

1. Deleted: engine, gearbox, clutch, fuel tank and lines, exhaust and after-treatment, starter, alternator, engine radiator. Added: battery pack and BMS, inverter, motor, reduction gear, OBC, DC-DC, HV junction box and harness, charging inlet, battery thermal loop and chiller, electric AC compressor, PTC heater or heat pump, electric vacuum or brake booster, HV labels and safety parts.
2. Single front or rear motor (not AWD): lowest cost and best efficiency; high utilisation favours efficiency over performance; FWD if converted from a petrol car, RWD on a dedicated platform.

16. Further learning

- Ehsani et al., *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles* (CRC Press): powertrain architecture and sizing.
- Larminie & Lowry, *Electric Vehicle Technology Explained* (Wiley).
- NPTEL, "Introduction to Hybrid and Electric Vehicles" (IIT Guwahati).

- Munro & Associates and similar teardown reports (e.g. Tesla Model 3/Y) for real component layouts.

Next: Module 2, electric traction motors.